



IMPROVING DIGITAL PRIVACY EDUCATION FOR CHILDREN THROUGH GAMING

SI 332: Privacy & Surveillance

Grace Ashworth, Eric Nielsen, Sarafina Chea



CLASS POLL:

**RAISE YOUR HAND IF YOUR PARENTS/GUARDIANS
TALKED TO YOU ABOUT HOW TO PROTECT YOUR
PRIVACY ONLINE.**



**IF YOU RAISED
YOUR HAND...**





OUR PROBLEM

“CHILDREN TODAY SPEND MUCH OF THEIR LIVES ONLINE BUT FEW RESOURCES TEACH THEM HOW TO PROTECT THEIR PERSONAL PRIVACY. GIVEN THE LACK OF MODERN EDUCATIONAL RESOURCES, CHILDREN ARE RARELY TAUGHT TOPICS SUCH AS DATA SAFETY, DIGITAL FOOTPRINTS, OR RESPONSIBLE INFORMATION SHARING. THIS LACK OF GUIDANCE LEAVES CHILDREN VULNERABLE TO PRIVACY RISKS AND UNSAFE ONLINE BEHAVIOR.”





CURRENT REGULATIONS ON CHILDREN'S PRIVACY IN THE UNITED STATES

CHILDREN'S ONLINE PRIVACY PROTECTION ACT (1998)

RESTRICTS HOW OPERATORS OF CHILD-DIRECTED WEBSITES AND ONLINE SERVICES COLLECT AND USE PERSONAL DATA FROM CHILDREN UNDER 13 (SANDOVAL, 2024).

Requirements under COPPA

- (1) Provide a clear privacy notice explaining data collection and use.
- (2) Directly notify parents about what data from the child is being collected, how it's used, and their rights.
- (3) Obtain verifiable parental consent before collecting personal data from children



COPPA FAILS TO ADDRESS THE FOLLOWING:

- **The monetization of children's data.**
- **Online predators creating accounts under false pretenses.**
- **Data breaches containing children's information.**

CHILDREN ARE INCREASINGLY BECOMING SURVEILED & DATAIFIED

From apps to everyday technologies...



- 92.8% of children aged 12-18 in western countries has access to the internet (Valcke, 2011).
- 68% of children under the age of 12 have their own smart-tablet (McClain et al., 2025).
- 40% of parents say their child 12 or under uses a voice assistant like Siri or Alexa (McClain et al., 2025).



OUR NEXT STEPS

As the world becomes increasingly digitalized, action must be taken to teach children how to protect themselves!

01

PUT EMPHASIS ON ONLINE SAFETY EDUCATION

02

MAKE ONLINE SAFETY EDUCATION ENJOYABLE FOR CHILDREN

03

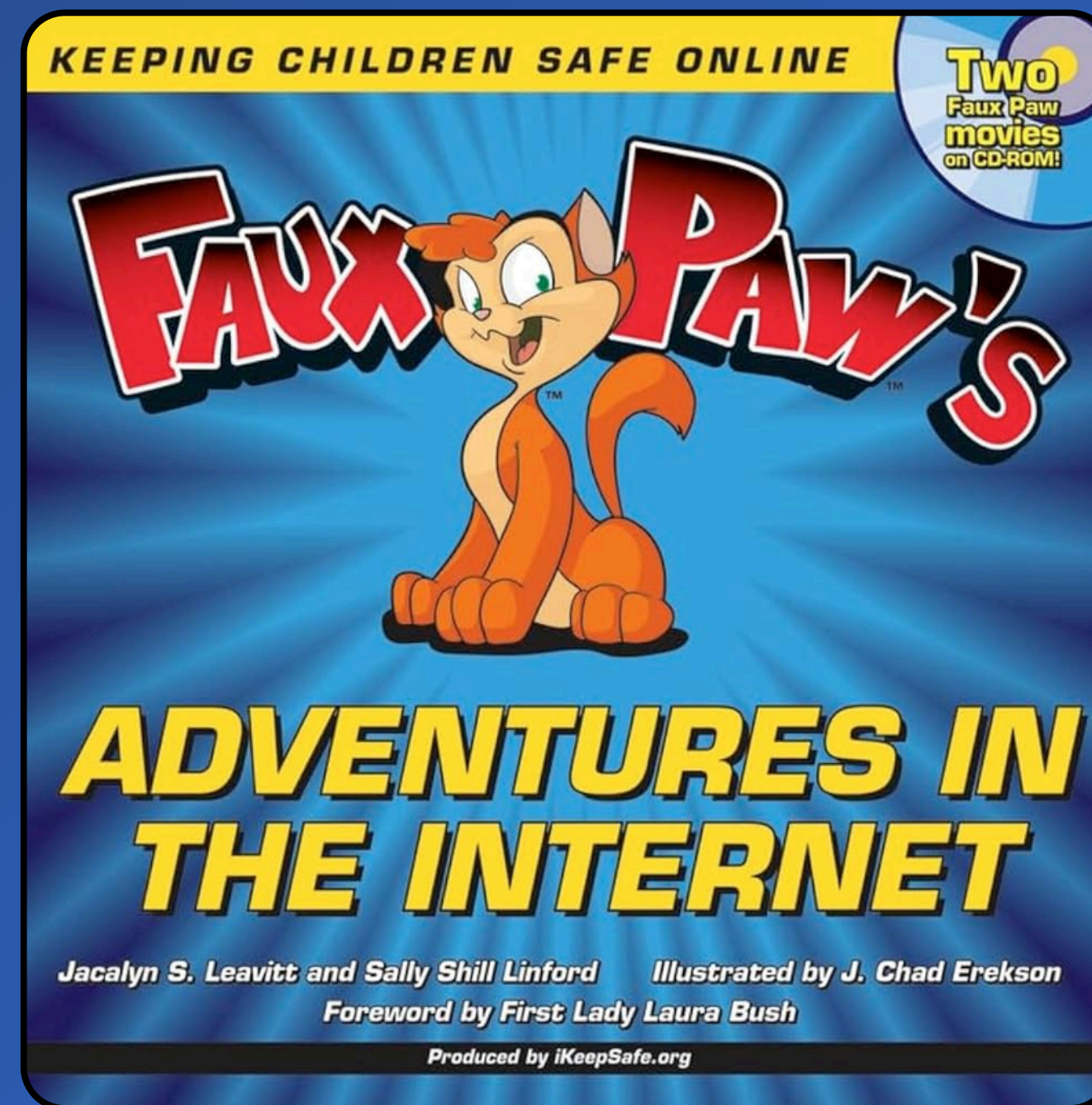
IDENTIFY PRIVACY AND SURVEILLANCE RISKS CHILDREN FACE

04

DETERMINE THE BEST WAY FOR CHILDREN TO LEARN AND APPLY THIS KNOWLEDGE



WE SAW THE SHORTCOMINGS OF CURRENT PROGRAMS WITH DR. GIROUARD-HALLAM'S LECTURE





POTENTIAL SOLUTION: AN EDUCATION PLAN

A digital safety education plan would focus on teaching children about data sharing, surveillance, and safe online behavior using real world scenarios. Teachers would guide students through safe decision making and what and what not to do.



**AGE APPROPRIATE
LESSONS**



**REAL WORLD
SCENARIOS**



**INTERACTIVE
ACTIVITIES**



**TEACHER
GUIDANCE**

HOWEVER, WE DECIDED AGAINST IT...

Despite the benefits of an educational plan, we ultimately decided that there are too many drawbacks.

➤ ACCESSIBILITY

Lesson plans can only be accessed in the classroom, limiting where & when children learn.

➤ ENGAGEMENT

Many children become bored or disinterested with standard education plan, limiting their learning.

➤ IMPLEMENTATION

Curriculum differs by school and state, making this difficult to implement on a large scale (Kumar & Hyde 2023).

➤ COST

The majority of lesson plans come at some cost, which would limit this to only the schools that could afford it.





IDENTIFYING GAPS



Online safety education options are available, but most of them fall short in providing meaningful and applicable skills that translate to the real world knowledge.

TOO GENERIC

Teaches vague messages (“Someone could be listening”) without explaining who the threat is or what the actual harm is.

LACKS HARM CONTEXT

Kids struggle to understand why privacy & surveillance matters and may misinterpret the risks.

FOCUSES ON “DON’TS”

Often tells kids what not to do instead of teaching concrete, positive behaviors.

OUR SOLUTION

A FREE-TO-PLAY & EASILY ACCESSIBLE MOBILE GAME THAT TEACHES ELEMENTARY TO MIDDLE SCHOOL CHILDREN ABOUT ONLINE RISKS AND NAVIGATING EVERYDAY SURVEILLANCE!



THE BENEFITS OF GAMING

- ▶ CHILDREN FIND VIDEO GAMES SIGNIFICANTLY MORE FUN THAN CLASSROOM LECTURES
- ▶ GAME-BASED LEARNING TOOLS PROVIDE STUDENTS WITH A SENSE OF AUTONOMY (ALOTAIBI, 2024)
- ▶ CHILDREN ARE FULLY IN CONTROL, ENHANCING PROBLEM SOLVING SKILLS AND CRITICAL THINKING (UNICEF, 2024)
- ▶ GAMES ARE WIDELY ACCESSIBLE TO CHILDREN, PARENTS/GUARDIANS, AND EDUCATORS





PRIVAPALS!



KEEP YOUR PRIVAPAL SAFE! LEARN WHEN, HOW, AND WHY TO PROTECT YOUR PERSONAL INFORMATION FROM SNEAKY CHARACTERS OR SYSTEMS THAT TRY TO WATCH, TRACK, OR COLLECT IT.

Navigating Privapals, children ages 9–13 will learn how to...

- Make smart online decisions
- Identify areas of surveillance
- Detect suspicious patterns
- Decline unnecessary cookies
- Manage privacy dynamically in various situations

PrivaPals

Start

New Pal

Settings





PrivaPals

\$ 350

1

Next Island

Berry

Hi, can I add you as a friend?

Yes - share public username

No Thanks



GAMING-SIMULTION EXAMPLES

Topic One: Data Safety

“Help your Privapal Stay Safe During Signup!”

- Kids decide what to enter in a fake signup form. Ted the safety dog will walk them through what makes a password safe.
- Oversharing leads to “Data Drip” and will show an animation of info leaking, visually showing kids how their data it is vulnerable to being exposed and seen by other



ENSURING COPPA-COMPLIANCE



**MINIMAL DATA
COLLECTION**



**PARENTAL
CONSENT**



**SAFE GAMEPLAY
& ANALYTICS**



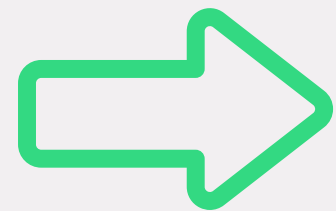
**DATA PROTECTION
& SECURITY**



MOVING FORWARD

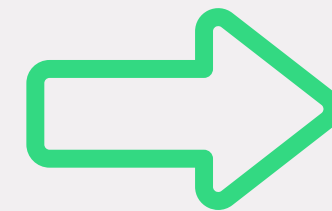
Finalize App Design & Content

Complete interactive game-play model, ensure tasks teach privacy and surveillance principles effectively.



Quality Assurance Testing

Test gameplay mechanics, check COPPA compliance features.



Conduct Beta Testing

Collecting feedback from both kids and parents.

THANK YOU!
QUESTIONS?



REFERENCES

Alotaibi, M. S. (2024). Game-based learning in early childhood education: a systematic review and meta-analysis. *Frontiers in Psychology, 15*, 1307881. <https://doi.org/10.3389/fpsyg.2024.1307881>

Finkelhor, D., Jones, L., & Mitchell, K. (2021). Teaching privacy: A flawed strategy for children's online safety. *Child Abuse & Neglect, 117*. <https://www.sciencedirect.com/science/article/pii/S014521342100137X#bbib0195>

Kumar, P. C., Chetty, M., Clegg, T. L., & Vitak, J. (2019). Privacy and Security Considerations For Digital Technology Use in Elementary Schools. *Conference on Human Factors in Computing Systems Proceedings*, 1–13. <https://doi.org/10.1145/3290605.3300537>

Kumar, P., & Hyde, L. (2023). EXPLORING HOW U.S. K-12 EDUCATION ADDRESSES PRIVACY LITERACY. *AoIR Selected Papers of Internet Research*. <https://doi.org/10.5210/spir.v2023i0.13439>

Mcclain, C., Faverio, M., Anderson, M., & Park, E. (2025, October). *How Parents Manage Screen Time for Kids*. Pew Research Center. <https://www.pewresearch.org/internet/2025/10/08/how-parents-manage-screen-time-for-kids/>

Sandoval, M. (2024, August). *Child Privacy in the Digital Era: Is COPPA Enough?* *Child Privacy in the Digital Era: Is COPPA Enough?* Digital Commons at American University College of Law. <https://digitalcommons.wcl.american.edu/cgi/viewcontent.cgi?article=1132&context=research>

Sinha, D. (2024, July). *COPPA Compliance: How to design COPPA compliant mobile apps for kids*. TechAhead. <https://www.techaheadcorp.com/blog/coppa-compliance/>

Valcke, M., De Wever, B., Van Keer, H., & Schellens, T. (1978). Long-term study of safe Internet use of young children. *Computers & Education, 57*(1). <https://doi.org/10.1016/j.compedu.2011.01.010>

Zhao, J., Wang, G., Dally, C., Slovak, P., Edbrooke-Childs, J., Van Kleek, M., & Shadbolt, N. (2019). 'I make up a silly name' Understanding Children's Perception of Privacy Risks Online. In *CHI Conference on Human Factors in Computing Systems Proceedings*, 1–13. <https://doi.org/10.1145/3290605.3300336>